

DSA 0604

Data Handling and Visualization

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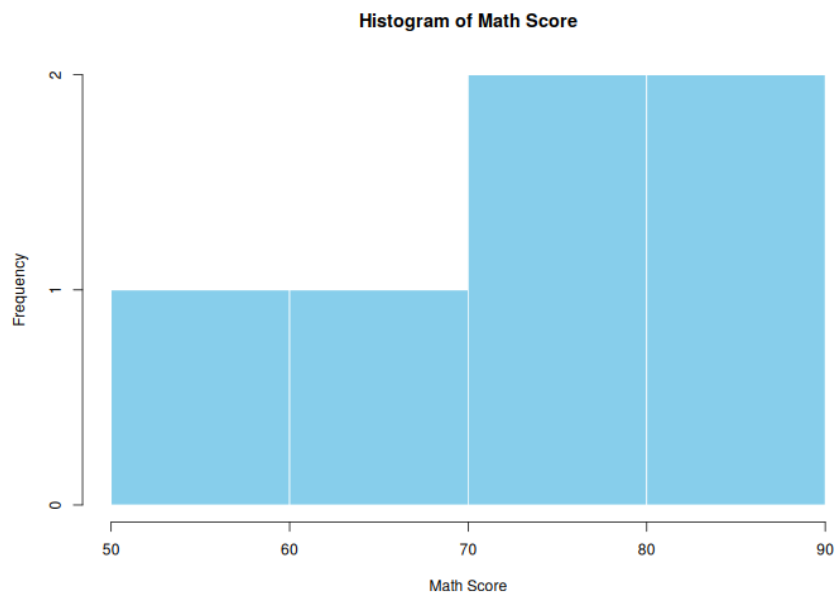
SET 26

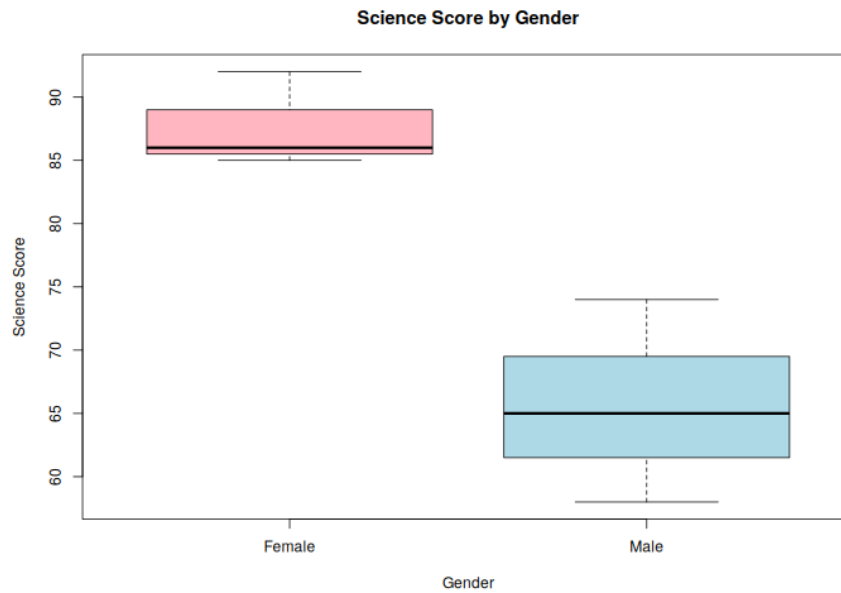
Dataset: Student_Mini_Data.csv

Student_ID	Gender	Study_Hours	Attendance	Math_Score	Science_Score	Exam_Date
S01	Male	2.0	78	62	65	2025-01-10
S02	Female	3.5	90	80	85	2025-01-10
S03	Male	1.5	70	55	58	2025-02-12
S04	Female	4.0	95	90	92	2025-02-12
S05	Male	2.8	85	72	74	2025-03-15
S06	Female	3.0	92	82	86	2025-03-15

1. Import the dataset in R. Plot a histogram for Math_Score. Draw a boxplot of Science_Score by Gender. Add proper title and labels. Interpret the results.

```
student_data <- data.frame(  
  Student_ID=c("S01", "S02", "S03", "S04", "S05", "S06"),  
  Gender=c("Male", "Female", "Male", "Female", "Male", "Female"),  
  Study_Hours=c(2.0, 3.5, 1.5, 4.0, 2.8, 3.0),  
  Attendance=c(78, 90, 70, 95, 85, 92),  
  Math_Score=c(62, 80, 55, 90, 72, 82),  
  Science_Score=c(65, 85, 58, 92, 74, 86),  
  Exam_Date=c("2025-01-10", "2025-01-10", "2025-02-12",  
    "2025-02-12", "2025-03-15", "2025-03-15"))  
  
hist(student_data$Math_Score, main="Histogram of Math Score",  
  xlab="Math Score", col="skyblue", border="white")  
  
boxplot(Science_Score ~ Gender, data=student_data,  
  main="Science Score by Gender", xlab="Gender",  
  ylab="Science Score", col=c("lightpink", "lightblue"))
```

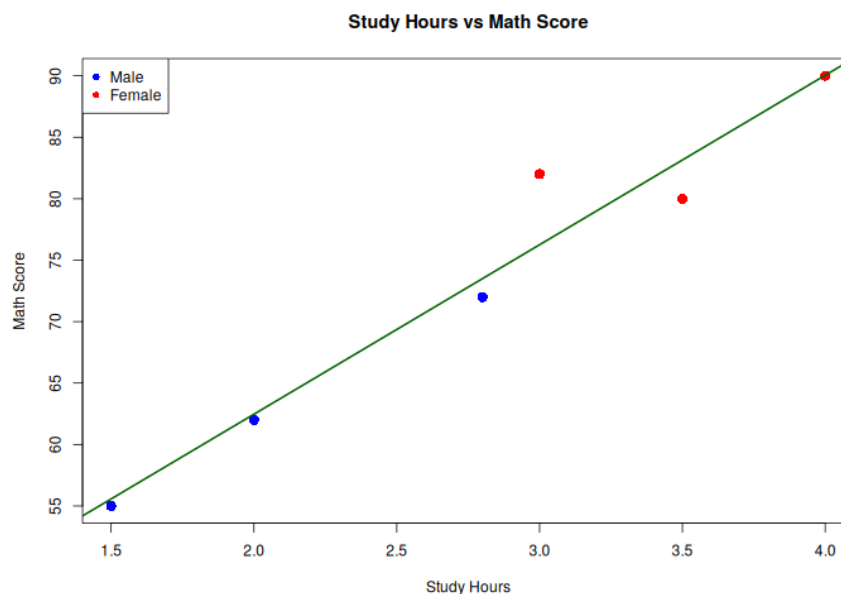




Interpretation: The histogram of Math_Score shows scores spread between 55 and 90 with no strong clustering, since the sample is small. The boxplot shows that Female students have a higher median Science_Score (about 85-86) than Male students (about 65-74), and the female group also shows a slightly wider spread, suggesting better and more variable performance among female students in this sample.

2. Create a scatter plot using R programming between Study_Hours and Math_Score. Use different colors for Gender. Add a regression line. Explain the relationship between study time and performance.

```
cols <- ifelse(student_data$Gender=="Male","blue","red")
plot(student_data$Study_Hours, student_data$Math_Score,
     col=cols, pch=16, cex=1.5,
     main="Study Hours vs Math Score",
     xlab="Study Hours", ylab="Math Score")
abline(lm(Math_Score ~ Study_Hours, data=student_data),
     col="darkgreen", lwd=2)
legend("topleft", legend=c("Male", "Female"),
     col=c("blue", "red"), pch=16)
```



Interpretation: The scatter plot shows a clear positive relationship between Study_Hours and Math_Score (correlation ≈ 0.97). As study hours increase, math scores rise almost linearly for both genders, and the fitted

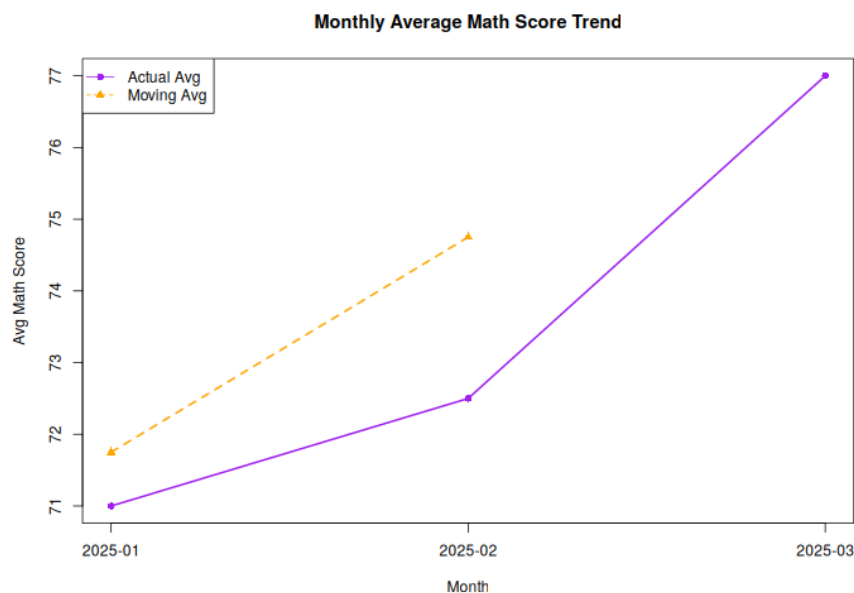
regression line confirms this strong positive trend - more study time is associated with better performance.

3. Using R, Convert Exam_Date into Date format. Calculate monthly average Math scores. Plot a line chart for the trend. Apply moving average smoothing. Identify any increasing or decreasing pattern.

```
student_data$Exam_Date <- as.Date(student_data$Exam_Date)
student_data$Month <- format(student_data$Exam_Date, "%Y-%m")
monthly_avg <- aggregate(Math_Score ~ Month, data=student_data, mean)

ma <- function(x, n=2) stats::filter(x, rep(1/n, n), sides=2)
monthly_avg$Smoothed <- as.numeric(ma(monthly_avg$Math_Score, 2))

plot(monthly_avg$Math_Score, type="o", pch=16, col="purple", lwd=2,
      main="Monthly Average Math Score Trend",
      xlab="Month", ylab="Avg Math Score")
lines(monthly_avg$Smoothed, type="o", pch=17, col="orange",
      lwd=2, lty=2)
legend("topleft", legend=c("Actual Avg", "Moving Avg"),
      col=c("purple", "orange"), lty=c(1, 2))
```



Interpretation: After converting Exam_Date and grouping by month, the average Math_Score rises steadily from January (71.0) to February (72.5) to March (77.0). The 2-point moving average smooths short-term fluctuation and confirms a consistent increasing trend across the three exam months.

4. Import Student_Mini_Data.csv into Tableau. Create a bar chart showing average Math scores by Gender. Create a line chart for score trends over time. Add a filter for Exam_Date. Design a simple interactive dashboard.

Tableau steps: (1) Connect to Student_Mini_Data.csv. (2) Drag Gender to Columns and AVG(Math_Score) to Rows, and change the mark type to Bar to get average Math score by gender. (3) On a new sheet, place Exam_Date on Columns and AVG(Math_Score) on Rows with mark type Line to show the score trend over time. (4) Add Exam_Date as a filter (quick filter, range/relative date) shown to the right of the view. (5) Combine both sheets on a new Dashboard, place the filter as a global control, and add titles so both views respond to the same Exam_Date filter interactively.

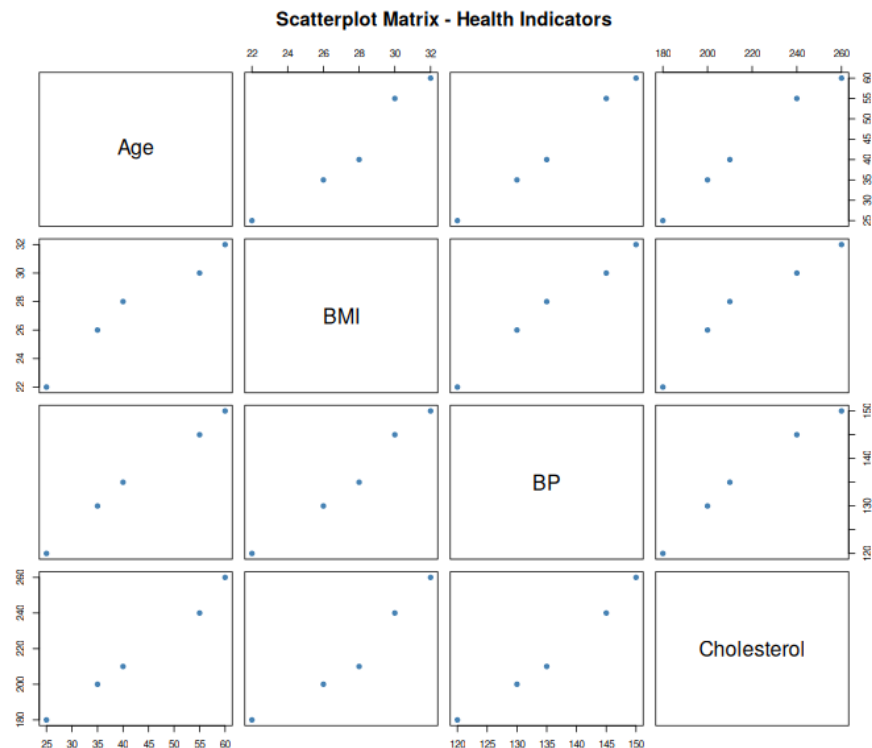
SET 27

Dataset: Patient Health Risk Analysis

Patient_ID	Age	BMI	BP	Cholesterol
P1	25	22	120	180
P2	40	28	135	210
P3	55	30	145	240
P4	35	26	130	200
P5	60	32	150	260

1. Using R, create a scatterplot matrix for Age, BMI, BP, and Cholesterol. Examine pairwise relationships to identify strong correlations, clusters, or potential risk patterns.

```
patient_data <- data.frame(  
  Patient_ID=c("P1", "P2", "P3", "P4", "P5"),  
  Age=c(25, 40, 55, 35, 60),  
  BMI=c(22, 28, 30, 26, 32),  
  BP=c(120, 135, 145, 130, 150),  
  Cholesterol=c(180, 210, 240, 200, 260))  
  
pairs(patient_data[,2:5],  
  main="Scatterplot Matrix - Health Indicators",  
  pch=19, col="steelblue")
```



Interpretation: All four variables move together almost perfectly (pairwise correlations of 0.97-1.00). As Age increases, BMI, BP and Cholesterol all increase in an almost linear fashion, indicating that older patients in this sample carry consistently higher risk across every indicator - a compounding risk pattern rather than an isolated one.

2. Construct Q-Q plot and ECDF for Cholesterol levels to assess distribution behaviour. Comment on normality assumption and skewness.

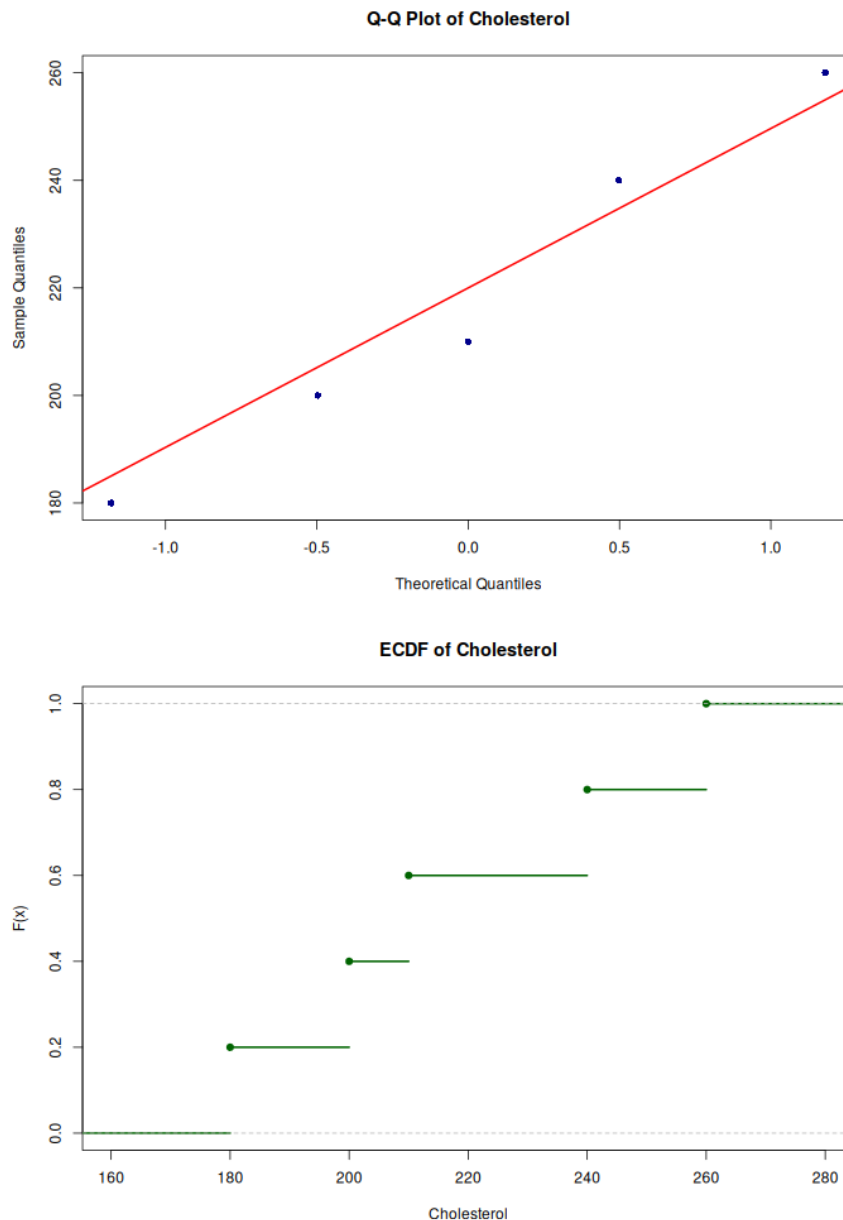
```
qqnorm(patient_data$Cholesterol, main="Q-Q Plot of Cholesterol",
```

```

pch=16, col="darkblue")
qqline(patient_data$Cholesterol, col="red", lwd=2)

plot(ecdf(patient_data$Cholesterol), main="ECDF of Cholesterol",
     xlab="Cholesterol", ylab="F(x)", col="darkgreen", lwd=2)

```



Interpretation: The Q-Q plot points lie reasonably close to the reference line, suggesting Cholesterol is roughly normally distributed for this small sample, though the top-end value (260) departs slightly, hinting at mild right skew. The ECDF rises in roughly even steps, consistent with values that are fairly evenly spread across the 180-260 range rather than clustered.

3. Using R, create a bar chart showing the average values of each health indicator (Age, BMI, BP, Cholesterol). Include axis labels, a descriptive title, and distinct colors for each bar. How do these averages relate to general patient health risk.

```

avg_values <- c(Age=mean(patient_data$Age),
               BMI=mean(patient_data$BMI),
               BP=mean(patient_data$BP),
               Cholesterol=mean(patient_data$Cholesterol))

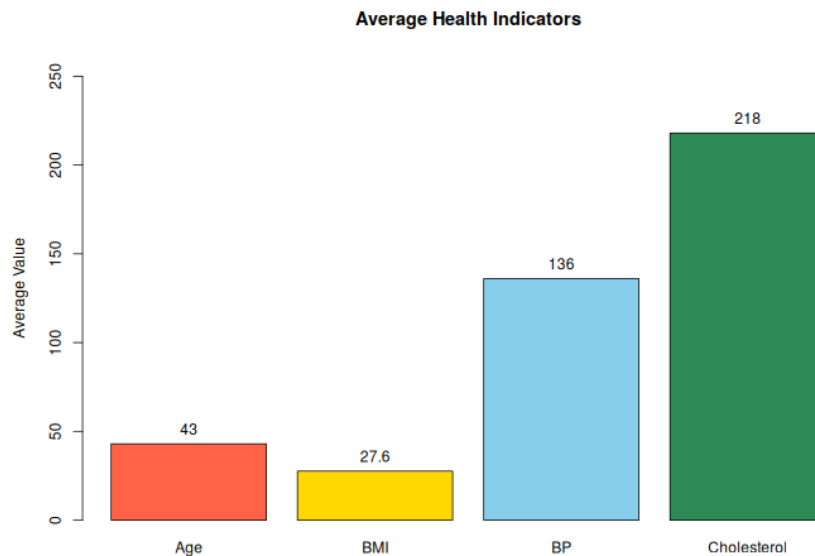
bp <- barplot(avg_values, main="Average Health Indicators",

```

```

ylab="Average Value",
col=c("tomato","gold","skyblue","seagreen"))
text(bp, avg_values, labels=round(avg_values,1), pos=3)

```



Interpretation: Average BP (136 mmHg) sits at the upper edge of the normal range and average Cholesterol (218 mg/dL) is in the borderline-high zone, while average BMI (27.6) falls in the overweight category. Together these averages suggest the overall patient group carries a moderate-to-elevated cardiovascular risk profile, driven mainly by BP and Cholesterol.

4. Create an interactive Tableau dashboard with a heatmap highlighting high-risk patients and an Age filter for exploring risk patterns. Include hover tooltips showing all health indicators. Analyze which patients are highest risk, how age affects risk, and how this dashboard can aid healthcare professionals in prioritizing care.

Tableau steps: (1) Connect to the Patient dataset. (2) Place Patient_ID on Rows and a risk measure (e.g. Cholesterol or a calculated Risk_Score) on Columns, set the mark type to Square/heatmap and color by the risk measure (red = high risk, green = low risk). (3) Add Age as a filter card so a user can restrict the view to a given age band. (4) Add all indicators (Age, BMI, BP, Cholesterol) to the Tooltip shelf so hovering over any cell reveals full patient detail. (5) Place the heatmap and filter on a Dashboard. Patient P5 (age 60) shows the darkest cell and is the highest-risk patient; risk clearly rises with age, which lets clinicians quickly triage older, higher-value patients for follow-up care.

SET 28

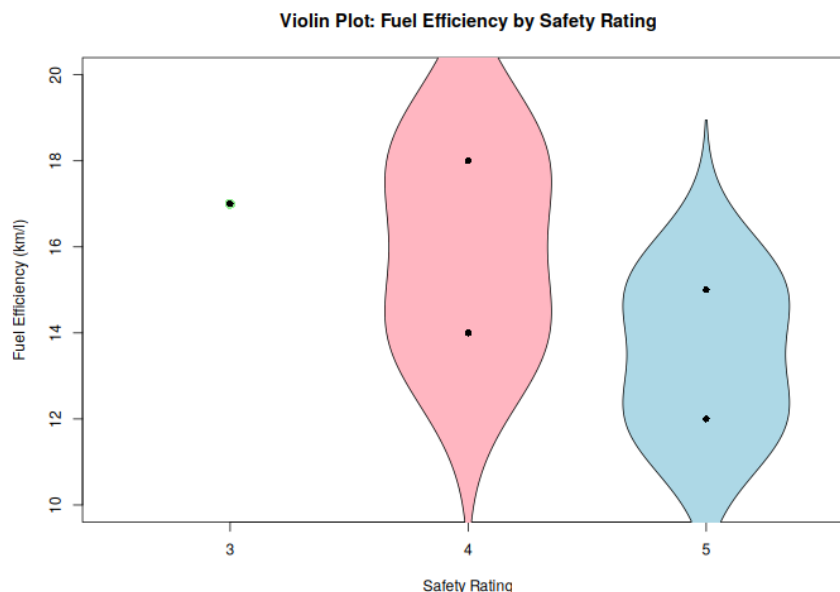
Dataset: Vehicle Performance Analysis

Vehicle_ID	Engine_Size (L)	Horsepower	Fuel_Efficiency (km/l)	Top_Speed (km/h)	Safety_Rating
V1	1.5	110	18	180	4
V2	2.0	150	15	200	5
V3	3.0	250	12	250	5
V4	2.5	200	14	220	4
V5	1.8	130	17	190	3

1. Using R, develop a violin plot to compare the distribution of Fuel_Efficiency across different Safety_Rating categories. Examine the spread, central tendency, and variability within each group, and interpret what this indicates about efficiency patterns.

```
vehicle_data <- data.frame(
  Vehicle_ID=c("V1", "V2", "V3", "V4", "V5"),
  Engine_Size=c(1.5, 2.0, 3.0, 2.5, 1.8),
  Horsepower=c(110, 150, 250, 200, 130),
  Fuel_Efficiency=c(18, 15, 12, 14, 17),
  Top_Speed=c(180, 200, 250, 220, 190),
  Safety_Rating=c(4, 5, 5, 4, 3))

# Violin plot (density-based, base R)
groups <- sort(unique(vehicle_data$Safety_Rating))
plot(NA, xlim=c(0.5, length(groups)+0.5),
     ylim=range(vehicle_data$Fuel_Efficiency)+c(-2, 2),
     xlab="Safety Rating", ylab="Fuel Efficiency (km/l)",
     main="Violin Plot: Fuel Efficiency by Safety Rating")
for (i in seq_along(groups)) {
  vals <- vehicle_data$Fuel_Efficiency[
    vehicle_data$Safety_Rating==groups[i]]
  d <- density(vals, adjust=1.5)
  w <- 0.35 * d$y/max(d$y)
  polygon(c(i-w, rev(i+w)), c(d$x, rev(d$x)), col="lightblue")
  points(rep(i, length(vals)), vals, pch=16)
}
```

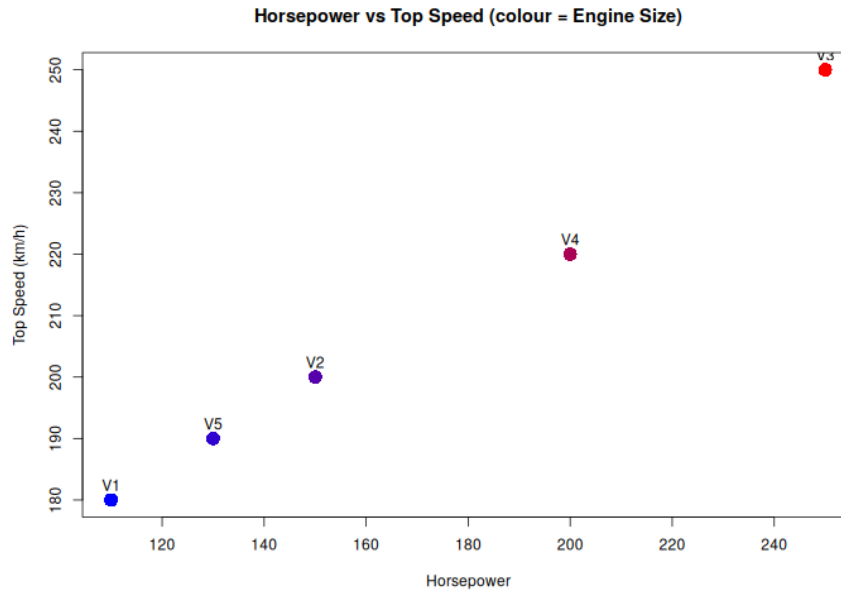


Interpretation: Safety_Rating 4 shows the widest spread in Fuel_Efficiency (14-18 km/l), Safety_Rating 5 clusters lower (12-15 km/l), and Safety_Rating 3 has a single high value (17 km/l). This suggests a general tendency for higher safety-rated vehicles to trade off some fuel efficiency, likely due to heavier safety equipment.

and more powerful engines.

2. Using R, construct a scatter plot to explore the relationship between Horsepower and Top_Speed, applying color differentiation based on Engine_Size. Analyze the strength and direction of correlation and find observable performance trends.

```
size_col <- colorRampPalette(c("blue","red"))(100)[
  as.numeric(cut(vehicle_data$Engine_Size, breaks=100))]
plot(vehicle_data$Horsepower, vehicle_data$Top_Speed,
     col=size_col, pch=16, cex=2,
     main="Horsepower vs Top Speed (colour = Engine Size)",
     xlab="Horsepower", ylab="Top Speed (km/h)")
text(vehicle_data$Horsepower, vehicle_data$Top_Speed,
     labels=vehicle_data$Vehicle_ID, pos=3)
```

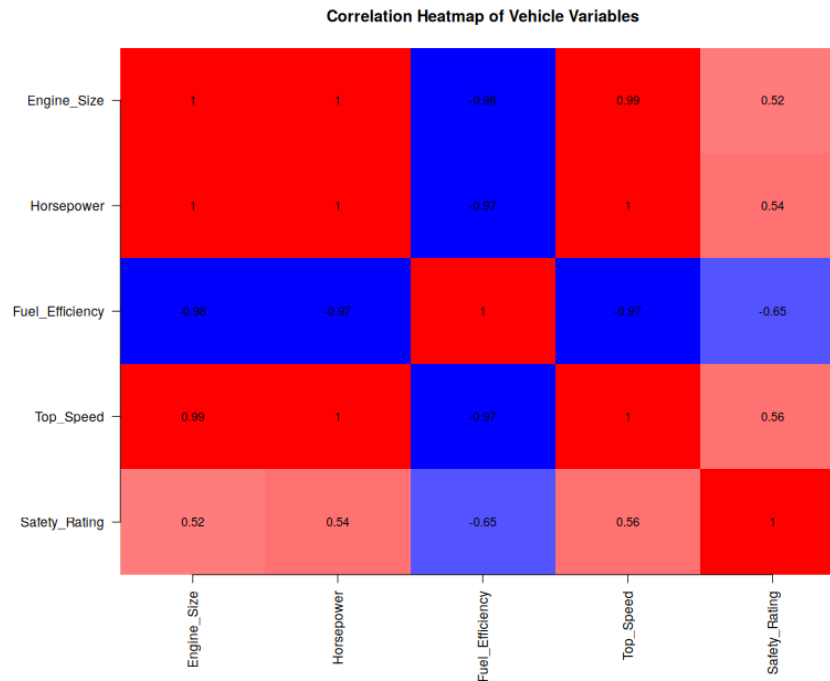


Interpretation: Horsepower and Top_Speed show a very strong positive correlation ($r \approx 1.00$). Points shift from blue (small engine) to red (large engine) as both Horsepower and Top_Speed rise, confirming that larger engine size drives higher horsepower, which in turn drives higher top speed.

3. Using R, create a correlation heatmap for all numerical variables in the dataset. Identify the variables most strongly associated with Top_Speed and their influence on overall vehicle performance.

```
num_data <- vehicle_data[,2:6]
corr_mat <- cor(num_data)

n <- ncol(corr_mat)
image(1:n, 1:n, corr_mat[,n:1], axes=FALSE, xlab="", ylab="",
     col=colorRampPalette(c("blue","white","red"))(50),
     main="Correlation Heatmap of Vehicle Variables")
axis(1, at=1:n, labels=colnames(corr_mat), las=2)
axis(2, at=1:n, labels=rev(colnames(corr_mat)), las=2)
for (i in 1:n) for (j in 1:n)
  text(i, n-j+1, round(corr_mat[i,j],2))
```

Interpretation: Top_Speed is almost perfectly correlated with Horsepower ($r=1.00$) and Engine_Size ($r=0.99$), and strongly negatively correlated with Fuel_Efficiency ($r=-0.97$). Safety_Rating has only a moderate relationship with Top_Speed ($r=0.56$). This confirms that engine size and horsepower are the dominant drivers of top speed, at the direct cost of fuel efficiency.

4. In Tableau, design an interactive dashboard incorporating bar charts and scatter plots, along with filters for Engine_Size and Safety_Rating, to highlight vehicles that balance fuel efficiency and high performance. Provide analytical interpretation of the insights derived from the dashboard.

Tableau steps: (1) Connect to the Vehicle dataset. (2) Build a bar chart of Vehicle_ID vs Fuel_Efficiency and a scatter plot of Horsepower vs Top_Speed (sized or colored by Fuel_Efficiency). (3) Add Engine_Size and Safety_Rating as filter cards so the user can narrow the view. (4) Place both charts on one Dashboard with the filters as global controls. V1 and V5 stand out as the best-balanced vehicles - decent top speed with the highest fuel efficiency - while V3 delivers the best raw performance at the largest efficiency cost, information that helps a buyer or fleet manager choose based on their priority.

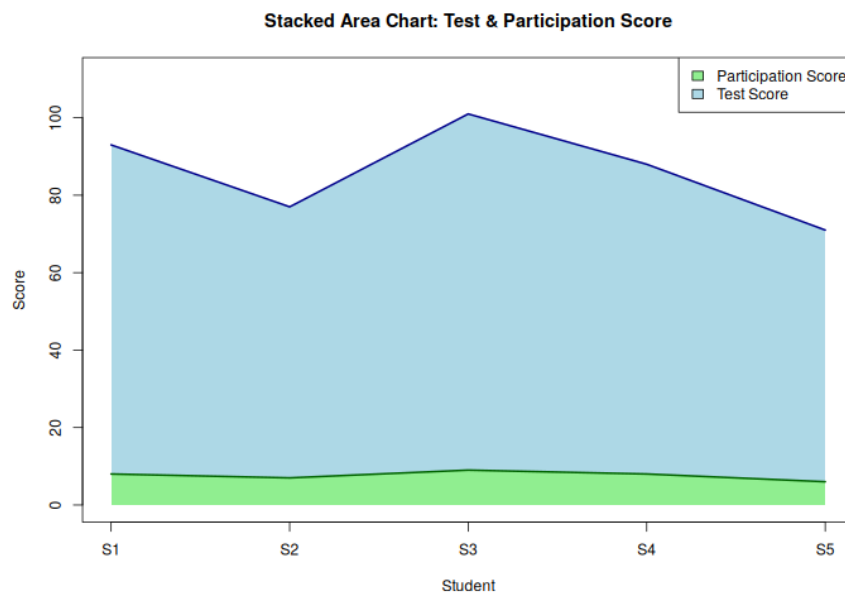
SET 29

Dataset: Student Academic Performance

Student_ID	Age	Study_Hours	Attendance (%)	Test_Score	Participation_Score
S1	19	12	90	85	8
S2	21	8	70	70	7
S3	20	15	95	92	9
S4	22	10	85	80	8
S5	23	7	60	65	6

1. Using R, create a stacked area chart showing Test_Score and Participation_Score across students. Analyze trends.

```
academic_data <- data.frame(  
  Student_ID=c("S1","S2","S3","S4","S5"),  
  Age=c(19,21,20,22,23),  
  Study_Hours=c(12,8,15,10,7),  
  Attendance=c(90,70,95,85,60),  
  Test_Score=c(85,70,92,80,65),  
  Participation_Score=c(8,7,9,8,6))  
  
x <- 1:nrow(academic_data)  
ts <- academic_data$Test_Score  
ps <- academic_data$Participation_Score  
plot(NA, xlim=c(1,length(x)), ylim=c(0,max(ts+ps)*1.1),  
  xlab="Student", ylab="Score",  
  main="Stacked Area Chart: Test & Participation Score")  
polygon(c(x,rev(x)), c(rep(0,length(x)),rev(ps)), col="lightgreen")  
polygon(c(x,rev(x)), c(ps,rev(ps+ts)), col="lightblue")  
legend("topright", legend=c("Participation Score","Test Score"),  
  fill=c("lightgreen","lightblue"))
```

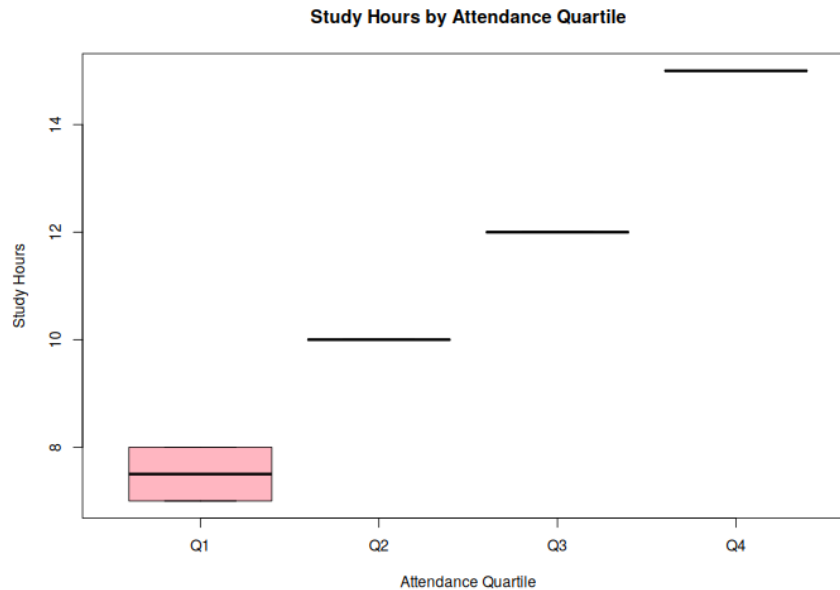


Interpretation: Both Test_Score and Participation_Score rise and fall together across students - S3 has the highest combined band and S5 the lowest - showing that students who participate more in class also tend to score higher on tests, a consistent positive relationship between engagement and performance.

2. Using R, generate a boxplot of Study_Hours grouped by Attendance quartiles. Interpret patterns. Include axis labels, title, and colors for each quartile. Analyze the boxplot to identify median and interquartile range of study hours per attendance quartile and any outliers in study hours.

```
academic_data$Attendance_Quartile <- cut(
  academic_data$Attendance,
  breaks=quantile(academic_data$Attendance, probs=seq(0,1,0.25)),
  include.lowest=TRUE, labels=c("Q1","Q2","Q3","Q4"))

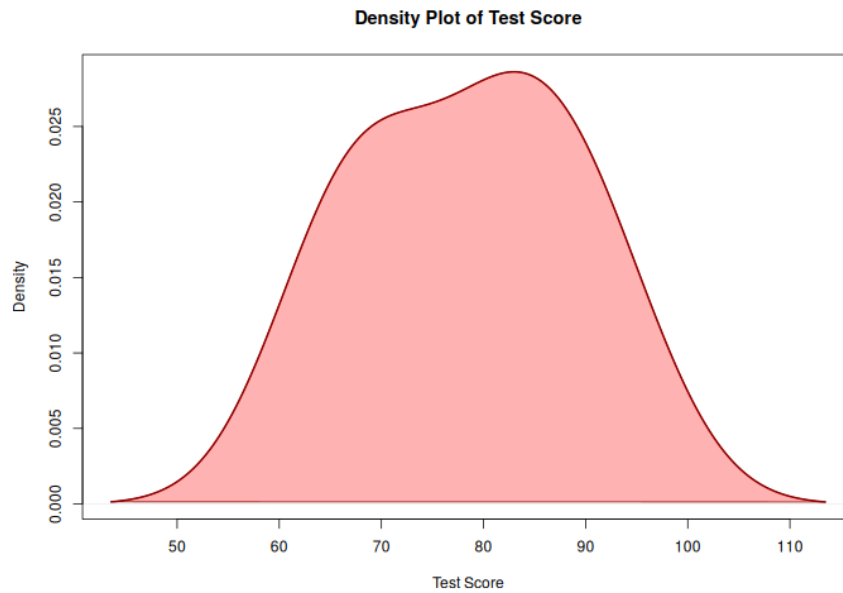
boxplot(Study_Hours ~ Attendance_Quartile, data=academic_data,
  main="Study Hours by Attendance Quartile",
  xlab="Attendance Quartile", ylab="Study Hours",
  col=c("lightpink","lightyellow","lightgreen","lightblue"))
```



Interpretation: Because each attendance quartile in this small dataset contains only a single student, each box collapses to a single value with no visible spread and no outliers (S5=7 hrs in Q1, S2=8 hrs in Q2, S1=12 hrs in Q3, S3=15 hrs in Q4). The pattern still shows a clear increasing trend: students with higher attendance quartiles tend to also log more study hours.

3. Using R, construct a density plot for Test_Score to evaluate score distribution and outliers.

```
plot(density(academic_data$Test_Score),
  main="Density Plot of Test Score",
  xlab="Test Score", col="darkred", lwd=2)
polygon(density(academic_data$Test_Score),
  col=rgb(1,0,0,0.3), border="darkred")
```



Interpretation: The density plot is fairly flat and spread across the 65-92 range with no sharp peak, reflecting the small, evenly-spread sample. There is no strong evidence of outliers - scores decline smoothly rather than showing an isolated spike.

4. In Tableau, design an interactive scatterplot with Study_Hours on X-axis, Test_Score on Y-axis, colored by Participation_Score. Include filters for Age.

Tableau steps: (1) Connect to the Student Academic Performance dataset. (2) Drag Study_Hours to Columns and Test_Score to Rows, set mark type to Circle. (3) Drag Participation_Score onto Color to shade each point by participation level. (4) Add Age to the Filters shelf and show it as a quick filter (slider or dropdown) so the view can be restricted by age. (5) Add tooltips with Student_ID for identification. The result is an interactive view where a viewer can isolate an age range and immediately see how study time, test performance and participation move together.

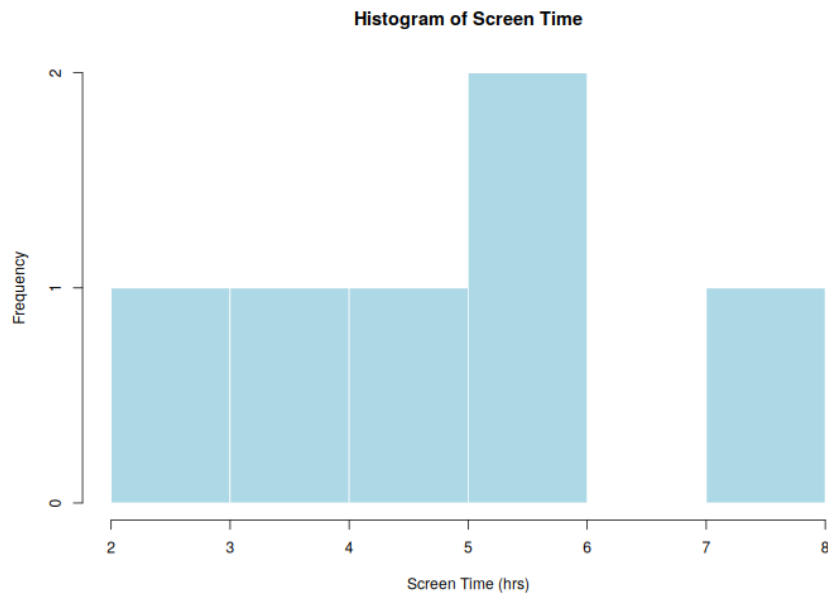
SET 30

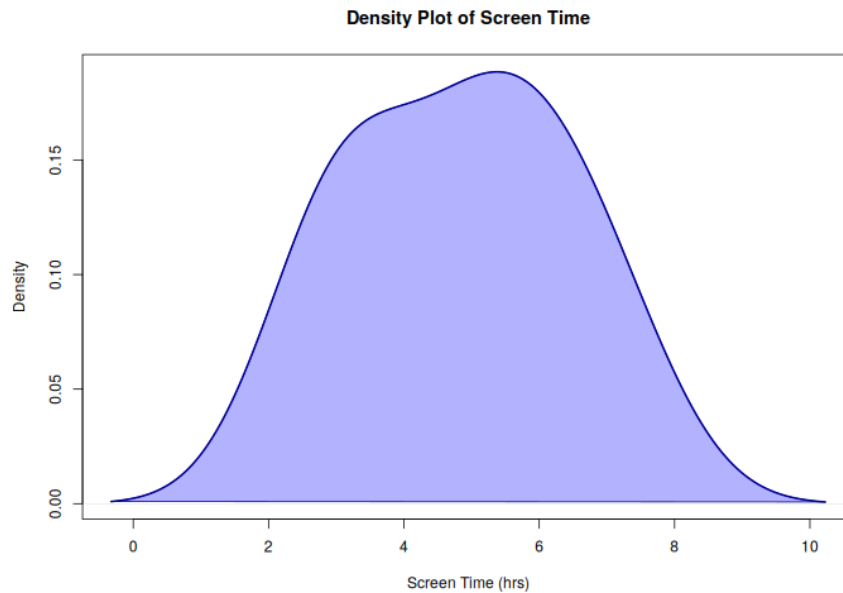
Dataset: Mobile_App_Usage.csv

User_ID	Gender	Age	Screen_Time (hrs)	App_Usage_Count	Data_Used (GB)	Satisfaction	Usage_Date
U01	Male	20	4.5	18	2.4	3	2025-01-08
U02	Female	22	6.0	25	3.8	5	2025-01-08
U03	Male	19	3.2	12	1.6	3	2025-02-11
U04	Female	21	7.1	30	4.5	5	2025-02-11
U05	Male	23	2.8	10	1.2	2	2025-03-14
U06	Female	20	5.4	22	3.1	4	2025-03-14

1. Import the dataset in R. Plot a histogram and density plot for Screen_Time. Add suitable colors and labels. Interpret the screen-time distribution.

```
app_data <- data.frame(  
  User_ID=c("U01", "U02", "U03", "U04", "U05", "U06"),  
  Gender=c("Male", "Female", "Male", "Female", "Male", "Female"),  
  Age=c(20, 22, 19, 21, 23, 20),  
  Screen_Time=c(4.5, 6.0, 3.2, 7.1, 2.8, 5.4),  
  App_Usage_Count=c(18, 25, 12, 30, 10, 22),  
  Data_Used=c(2.4, 3.8, 1.6, 4.5, 1.2, 3.1),  
  Satisfaction=c(3, 5, 3, 5, 2, 4),  
  Usage_Date=c("2025-01-08", "2025-01-08", "2025-02-11",  
               "2025-02-11", "2025-03-14", "2025-03-14"))  
  
hist(app_data$Screen_Time, main="Histogram of Screen Time",  
     xlab="Screen Time (hrs)", col="lightblue", border="white")  
  
plot(density(app_data$Screen_Time),  
     main="Density Plot of Screen Time",  
     xlab="Screen Time (hrs)", col="darkblue", lwd=2)  
polygon(density(app_data$Screen_Time),  
       col=rgb(0,0,1,0.3), border="darkblue")
```



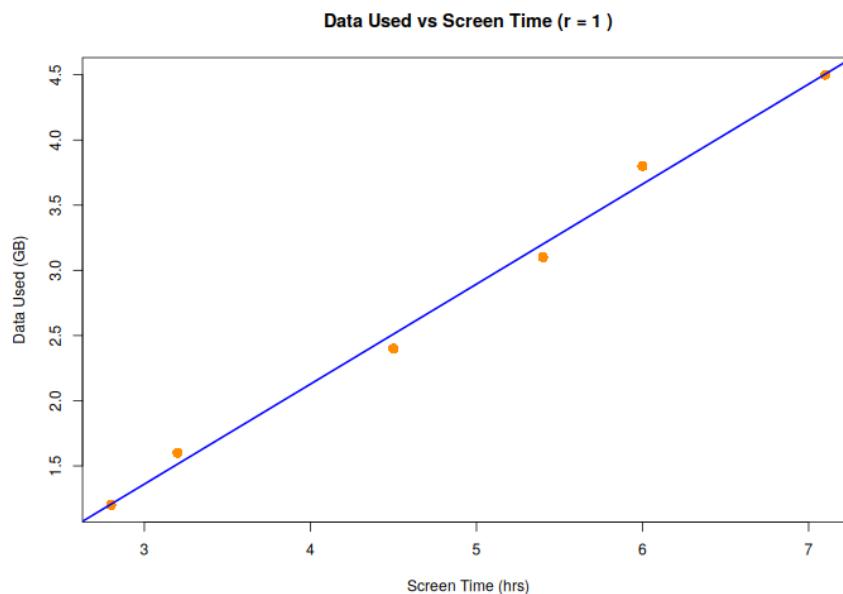


Interpretation: Screen_Time ranges from about 2.8 to 7.1 hours with values spread fairly evenly across that range. The density plot shows a broad, gently rounded shape rather than a sharp peak, indicating no single dominant usage pattern - users vary considerably in daily screen time.

2. Create a scatter plot using R programming between Data_Used and Screen_Time. Calculate the correlation between them. Add a trend line. Explain the relationship.

```
correlation <- cor(app_data$Screen_Time, app_data$Data_Used)

plot(app_data$Screen_Time, app_data$Data_Used, pch=16,
     col="darkorange", cex=1.5,
     main=paste("Data Used vs Screen Time (r =",
               round(correlation,2), ")"),
     xlab="Screen Time (hrs)", ylab="Data Used (GB)")
abline(lm(Data_Used ~ Screen_Time, data=app_data),
      col="blue", lwd=2)
```

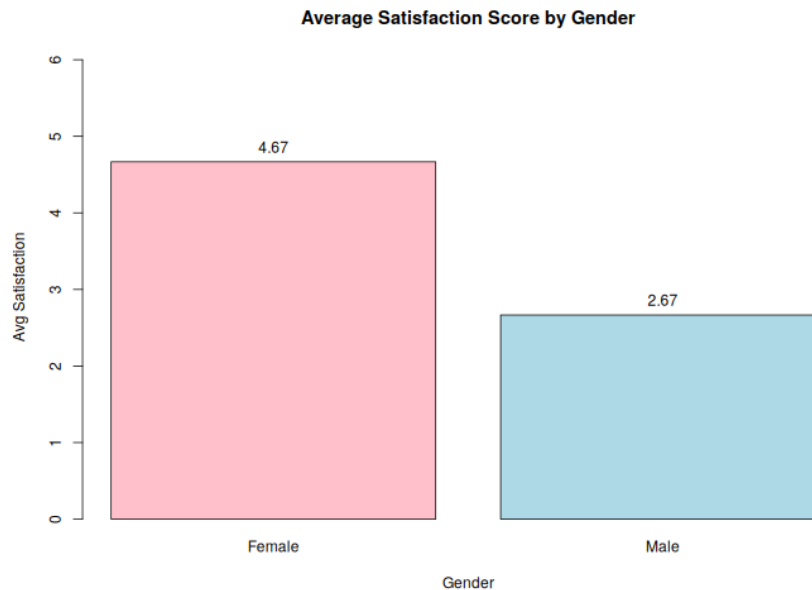


Interpretation: Data_Used and Screen_Time are very strongly positively correlated ($r \approx 0.997$). The points fall almost exactly on the fitted trend line, confirming that data consumption scales directly with the amount of time spent on the screen - roughly 0.6 GB of data per additional hour of screen time in this sample.

3. Using R, find the average Satisfaction score for each Gender. Create a bar chart for the averages. Add value labels. Interpret user satisfaction patterns.

```
avg_sat <- aggregate(Satisfaction ~ Gender, data=app_data, mean)

bp <- barplot(avg_sat$Satisfaction, names.arg=avg_sat$Gender,
              col=c("pink","lightblue"),
              main="Average Satisfaction Score by Gender",
              xlab="Gender", ylab="Avg Satisfaction")
text(bp, avg_sat$Satisfaction, labels=round(avg_sat$Satisfaction,2),
     pos=3)
```



Interpretation: Female users report a noticeably higher average satisfaction (4.67) than male users (2.67). Since female users in this sample also show higher Screen_Time and App_Usage_Count, greater engagement with the app appears to be associated with higher satisfaction.

4. Import Mobile_App_Usage.csv into Tableau. Create a bubble chart (Screen Time vs Data Used). Use color for Gender and size for App Usage Count. Create a line chart for usage trends. Add filters for Gender and Usage_Date. Design a mini dashboard.

Tableau steps: (1) Connect to Mobile_App_Usage.csv. (2) Place Screen_Time on Columns and Data_Used on Rows, set mark type to Circle; drag Gender to Color and App_Usage_Count to Size to build the bubble chart. (3) On a new sheet, place Usage_Date on Columns and a usage measure (e.g. SUM(App_Usage_Count) or AVG(Screen_Time)) on Rows with mark type Line for the usage trend. (4) Add Gender and Usage_Date to the Filters shelf as quick filters. (5) Combine the bubble chart and the trend line on one Dashboard with shared filters to create an interactive mini dashboard for exploring usage behaviour.